

Aaron Emmanuel

Mechatronics | Robotics & Systems Integration

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TECHNICAL EXPERIENCE

Associate Analyst | Rotational Engineering Program

May 2025 - Present

Kinectrics

- **Inspection and Maintenance Systems (Sep 2026 - Present):** Coordinate component needs with project teams, vendors and machine shops; assemble and functionally test purchased parts before final integration, including work on Calandria Relief System Modifications.
- **Technology and Strategic Ventures (Jan - Sep 2026):** Led proposals and coordinated technical input for radioisotope power and heating applications, contributing to Kinectrics' first Canadian Space Agency partnership.
- **Design and Engineering Services (May 2025 - Jan 2026):** Performed mechanical design analyses and validated electrical/instrumentation-and-control requirements to support qualification of replacement motors, relief valves and actuators at Bruce A and B nuclear stations.

Technical Lead | Systems Design and Integration

Jun 2024 - Present

Circulr

- Designed, built, programmed and deployed a compact reverse-vending machine for reusable jars and beverage cans as Circulr's sole technical contributor, bringing the system into store testing at Fiesta Farms.
- Integrated mechanical handling, electrical assembly and Raspberry Pi/Python controls with barcode validation, refund values and backend records. Processed approximately one item per second across six jar types and beverage cans, correctly accepting or rejecting every container in the evaluated test set.
- Designed a sensor-controlled storage elevator that moved with fill level to limit glass-container drop height and increase storage capacity.
- Enabled automatic recovery from momentary power loss and Raspberry Pi freezes by implementing boot-time autostart, watchdog monitoring and automatic application restart; the machine remained in service with no reported failures for six months after March 2026.

Co-Founder and Systems Lead | Underwater Robotics

Apr 2026 - Present

MAKO Technologies

- Co-founded MAKO Technologies to develop a clean-sheet underwater inspection crawler for Clearwater, with scope spanning mechanical, hydraulic, electrical, sensing, controls and software.
- Completed customer requirements and market research; now selecting components and developing a digital model around the competing demands of fitting through a manhole and producing sufficient power and traction to operate against water flow.

Robotic Systems Designer | Research and Development

Jan 2024 - Aug 2025

Clearwater Structures

- Rebuilt the electronics, control software and communications of an inoperable underwater inspection crawler, restoring full-system operation and demonstrating it underwater at a customer job site.
- Developed custom Python controller software linking onboard and truck-based Raspberry Pis by local IP over Ethernet/fibre, enabling remote operation and automatic stopping on communication loss; integrated a physical emergency stop.
- Integrated pan-tilt-zoom CCTV, LiDAR and an IMU into the rebuilt control system to expand inspection visibility and motion sensing.

SELECTED TECHNICAL PROJECTS

Collaborative Autonomous Robot Delivery System

2024 - 2025

Ontario Tech University | Undergraduate capstone

- Wrote and physically tuned a PID balance controller using IMU feedback and ODrive motor telemetry for a two-robot delivery platform.
- Programmed and integrated the electrical/control stack using Jetson, Teensy, ODrive and CAN; wired and assembled hardware and tested subsystem operation on the robots.
- Integrated a spring-assisted hip/leg mechanism that enabled the robots to stand after the existing motors proved unable to lift the system.

ASL Recognition Glove | Wearable Sensing

Dec 2025

University of Toronto | MEng course project

- Delivered a working sign-language recognition prototype in two weeks, owning data collection, evaluation of four classifiers, selection of K-nearest neighbours and a live webpage with spoken-letter output.
- Built an eight-input Bluetooth pipeline from an ESP32 to a laptop using five flex sensors and the MPU6050's three acceleration axes, recognizing all 26 American Sign Language alphabet letters, including moving gestures J and Z.
- Achieved 99.6% held-out accuracy on a two-user dataset with hundreds of samples per letter and an 80/20 train/test split, then demonstrated the system through live recognition.

TECHNICAL SKILLS

Programming and controls: Python, embedded programming, PID control, machine-learning classification.

Embedded hardware and interfaces: Raspberry Pi, ESP32, Jetson, Teensy, ODrive, CAN, Ethernet/fibre and Bluetooth.

CAD and electronics design: SolidWorks, Siemens NX, Autodesk Fusion and KiCad; parts, assemblies, drawings, structural FEA, motion simulation and schematic/PCB design.

Fabrication and test: Sensor integration, wiring harnesses, soldering, crimping, 3D printing, multimeter/oscilloscope testing and field troubleshooting.

EDUCATION

Master of Engineering | Mechanical and Industrial Engineering

2025 - 2027

University of Toronto

- Robotics Emphasis; degree in progress.

Bachelor of Engineering | Mechatronics Engineering

2020 - 2025

Ontario Tech University

- Graduated with Highest Distinction.